

Electrodynamics and Relativity

Problem 12.1

Problem 12.2

Problem 12.3

Problem 12.4

As the outlaws escape in their getaway car, which goes $\frac{3}{4}c$, the police officer fires a bullet from the pursuit car, which only goes $\frac{1}{2}c$. The muzzle velocity of the bullet (relative to the gun) is $\frac{1}{3}c$. Does the bullet reach its target (a) according to Galileo, (b) according to Einstein?

Solution:

(a) Adding up the velocities with Galileo's velocity addition rule, the resulting velocity of the bullet is

$$\frac{1}{2}c + \frac{1}{3}c = \frac{5}{6}c > \frac{3}{4}c$$

so the bullet does reach its target.

(b) Using Einstein's velocity addition rule, the result is instead

$$v_{\text{bullet}} = \frac{v_{\text{car}} + v_{\text{muzzle}}}{1 + \frac{v_{\text{car}} v_{\text{muzzle}}}{c^2}} = \frac{\frac{5}{6}c}{1 + \frac{1}{6}} = \frac{5}{7}c < \frac{3}{4}c$$

so the bullet does not reach its target.

Problem 12.5**Problem 12.6****Problem 12.7****Problem 12.8****Problem 12.9**

A Lincoln Continental is twice as long as a VW Beetle, when they are at rest. As the Continental overtakes the VW, going through a speed trap, a (stationary) policeman observes that they both have the same length. The VW is going at half the speed of light. How fast is the Lincoln going?

Solution:

The length of the cars while stationary is denoted L . With length contraction due to the velocity accounted for the length is denoted L' . With v_{vw} given and $L_{lc} = 2L_{vw}$, the solution is obtained by setting $L'_{lc} = L'_{vw}$ and solving for v_{lc} .

$$\begin{aligned} L_{lc} &= 2L_{vw} \\ L'_{vw} &= L_{vw} \sqrt{1 - \frac{v_{vw}^2}{c^2}} = L_{vw} \frac{\sqrt{3}}{2} \\ L'_{vw} &= L'_{lc} = L_{lc} \sqrt{1 - \frac{v_{lc}^2}{c^2}} = L_{vw} \frac{\sqrt{3}}{2} \end{aligned}$$

Using $L_{lc} = 2L_{vw}$ and solving for v_{lc} , the velocity for which the length of the Lincoln Continental equals the length of the VW Beetle is

$$v_{lc} = \frac{\sqrt{13}}{4}c$$

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